I used the modified template to create a TestStack and then also forked the python-project repository to my own repository. That is what was being monitored. https://github.com/timbertrail9/python-project.

Environment

- AWS Region: US East (N. Virginia), `us-east-1`.
- Stack name: [TraceTestStack]

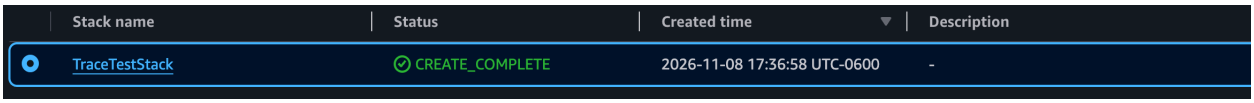
- Pipeline name: [TraceTestStack-AppCodePipeline-ECvoZTUCE4Ib]
  - Had some troubles successfully using connection to GitHub, which caused me to have to re-create the stack a few times. Then I was able to get the build to complete.
- Monitored GitHub repository and branch: [https://github.com/timbertrail9/python-project.git]
- Submitted GitHub repository URL: [https://github.com/timbertrail9/CodePipelineHW.git]

Verification results

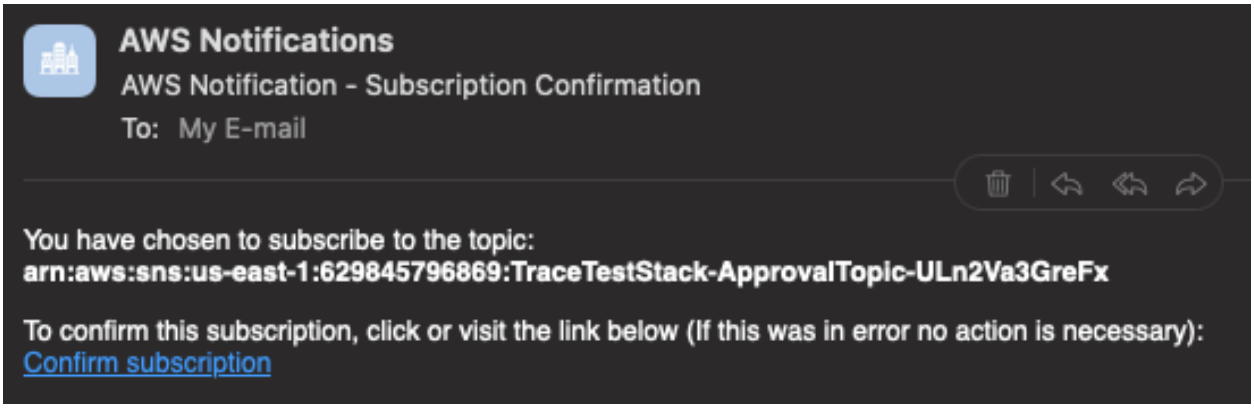
1. Stack deployment —

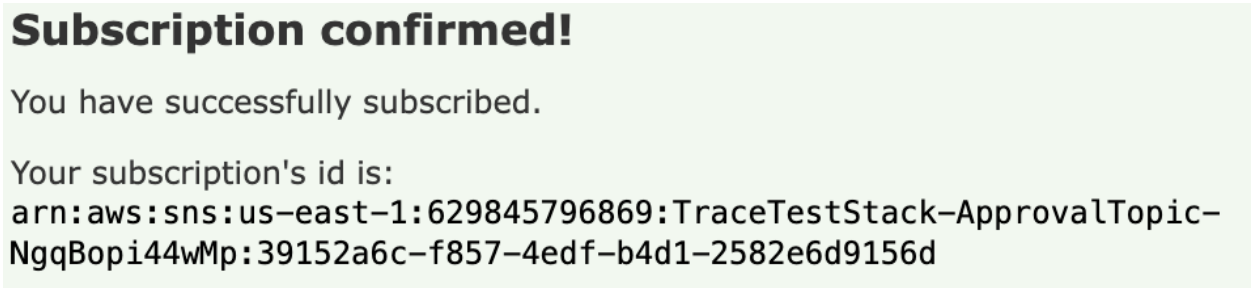
I deployed the CloudFormation template and observed that the stack completed successfully. The original launch-configuration error was resolved by using an EC2 launch template for the existing Auto Scaling group.

Evidence: CloudFormation stack status



2. Email subscription — completed



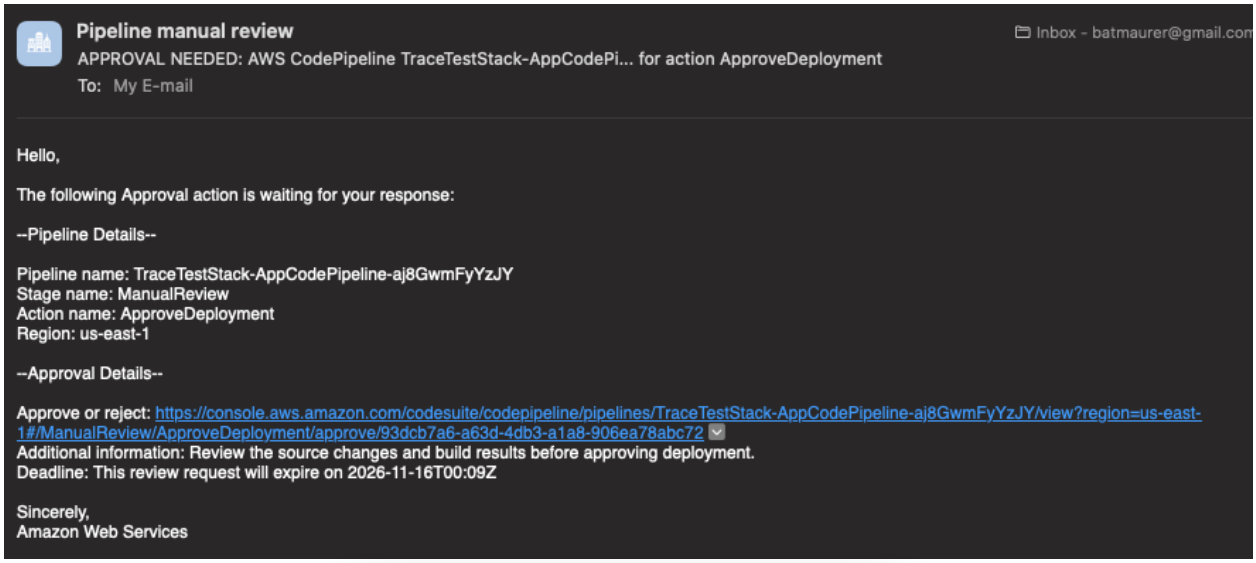


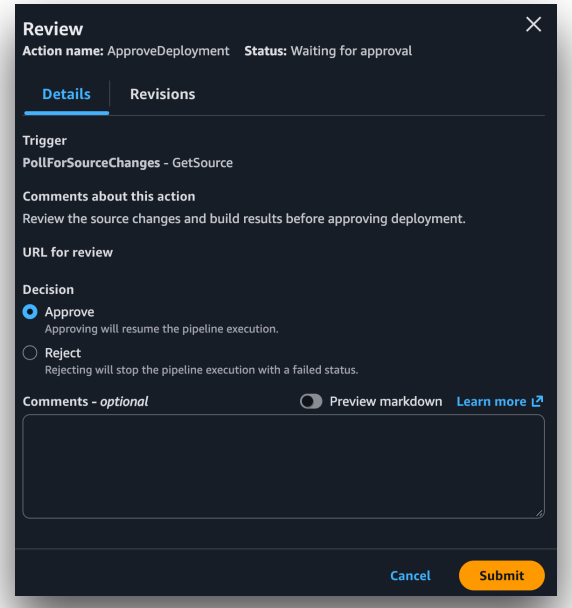
I opened the Amazon SNS subscription confirmation email and accepted the subscription. This enables the address to receive messages published to the approval topic. This confirmation is separate from approving a pipeline execution.

3. GitHub change and automatic trigger, build, and manual approval

Make a small change to the README file. Confirm that an e-mail is sent and that I can approve it.

Had to check to make sure pollforsourcechanges was set to true.





4. Testing and Deployment

Testing the build and fixing errors
The Python build initially failed while collecting the application test. The log showed:
ImportError: cannot import name 'url_quote' from 'werkzeug.urls'
Phase complete: PRE_BUILD State: FAILED

The project used Flask 2.0.3, but Werkzeug was not pinned to a compatible version. I added Werkzeug==2.0.3 to requirements.txt. I also removed the separate pytest upgrade from buildspec.yml so the build would use the version listed in the requirements. On the next attempt, PRE_BUILD succeeded.
The build then failed when compiling the CSS. The log showed that Less 4.9.1 required Node 18 or newer, but the build environment used Node 14.21.3. I changed the install command to:
npm install -g less@4.2.0

After these corrections were committed to the source repository, the build succeeded.

Build run	Status	Build number	Source version	Submitter	Duration	Completed
AppBuildProject-MrG1S8yDrod0:a4e4ec35-ab26-4a03-974f-2654643e39a4	Succeeded	2	arn:aws:s3:::test-artifactbucket-ziaspjytb1az/test-AppCodePipeline/SourceCode/RCJH8Nn.zip	codepipeline/test-AppCodePipeline-E3PCdoC28gkG	50 seconds	4 minutes ago
AppBuildProject-MrG1S8yDrod0:7e65fb72-aede-4426-9acd-2b7e652a5529	Succeeded	1	arn:aws:s3:::test-artifactbucket-ziaspjytb1az/test-AppCodePipeline/SourceCode/Irnz5aO.zip	codepipeline/test-AppCodePipeline-E3PCdoC28gkG	46 seconds	22 minutes ago

Testing manual approval and deployment

The manual approval action succeeded, and the pipeline continued to deployment. This showed that approving the action allowed the pipeline to proceed to the next stage.
The deployment uses CodeDeployDefault.OneAtATime. I watched the first instance complete, then saw the second instance go through BlockTraffic before continuing with its update. The application installed successfully on both instances. This verified that deployment continued after approval and completed on both servers.

Cleanup: Deleted the AWS Stack.

Instance ID	Duration	Status	Most recent event	Error code
i-08389501bac85fd71	5 minutes 49 seconds	Succeeded	AfterAllowTraffic	
BeforeBlockTraffic	less than one second	Succeeded		
BlockTraffic	5 minutes 13 seconds	Succeeded		
AfterBlockTraffic	less than one second	Succeeded		
ApplicationStop	less than one second	Succeeded		
DownloadBundle	less than one second	Succeeded		
BeforeInstall	less than one second	Succeeded		
Install	less than one second	Succeeded		
AfterInstall	1 second	Succeeded		
ApplicationStart	less than one second	Succeeded		
ValidateService	less than one second	Succeeded		
BeforeAllowTraffic	less than one second	Succeeded		
AllowTraffic	25 seconds	Succeeded		
AfterAllowTraffic	less than one second	Succeeded		
i-0d0ca197ae8d9dbde	-	In progress	BlockTraffic	
BeforeBlockTraffic	less than one second	Succeeded		
BlockTraffic	-	In progress		
AfterBlockTraffic	-	Pending		